STUDENT RETENTION & ACADEMIC PERFORMANCE

An interactive analysis of academic outcomes, performance gaps and student support priorities

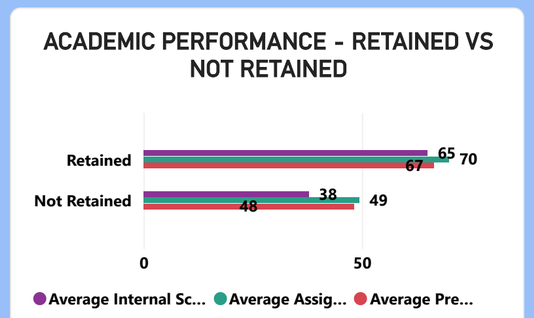
Department
- Commerce
- Computer Applications
- Humanities
- Management
- Science

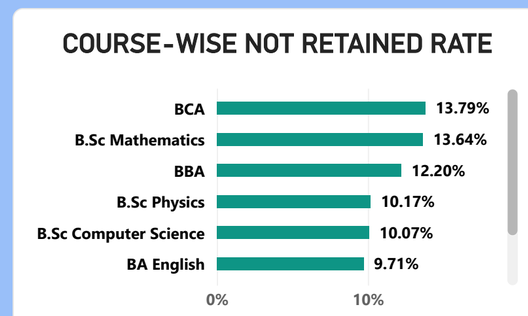
State
- Karnataka
- Kerala
- Maharashtra
- Tamil Nadu
- Telangana
- Unknown

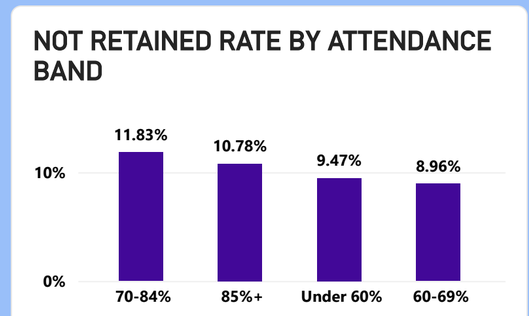
Course
- B.Com
- B.Sc Computer Science
- B.Sc Mathematics
- B.Sc Physics
- BA Economics
- BA English
- BBA
- BCA

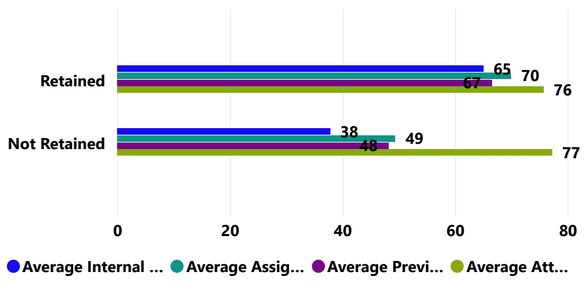
Gender
- Female
- Male

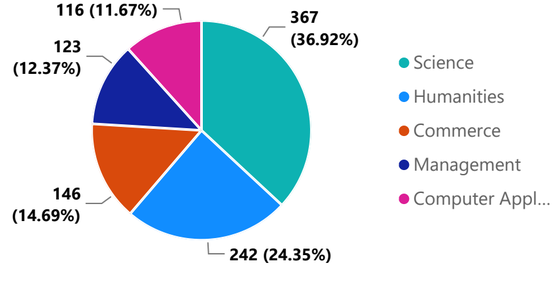
Total Students	Not Promoted Students	Average Final Score	Promotion Rate
994	104	64.83	89.54%

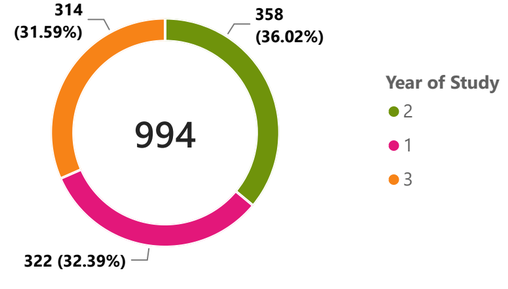












STUDENT RETENTION: EXECUTIVE NARRATIVE & KEY INSIGHTS

A concise overview of the dataset, dashboard purpose, and analytical approach.

EXECUTIVE NARRATIVE

A concise overview of the dataset, dashboard purpose, and analytical approach.

The Student Retention Dashboard analyses 994 student records across five departments, eight courses, and three years of study. It provides an overview of academic outcomes through four key performance indicators: total students, students not promoted, average final score, and promotion rate.
Interactive filters for department, course, gender, and state allow users to explore differences across student groups. The report compares average Internal Exam, Assignment, and Previous Exam scores between promoted and non-promoted students, examines non-promotion rates across courses and attendance bands, and presents student distribution by department and year of study.
The analysis identifies clear differences in assessment performance between students who are promoted and those who are not. It also examines whether attendance patterns distinguish these groups and highlights courses that may warrant closer academic review.
The primary objective is to support evidence-based decisions, identify students who may benefit from additional academic assistance, and recommend targeted interventions. The findings describe patterns within this dataset and should not be interpreted as proof of causation.

Classification note: In this report, “Not Retained” groups students whose Academic Status is “Needs Support” or “Repeat.” It does not mean that these students have dropped out.

KEY INSIGHTS

KEY INSIGHTS — WHAT THE DATA REVEALS

01. Student Retention and Promotion Outcomes

The dataset contains 994 students, of whom 890 are promoted and 104 are not promoted. The non-promoted group comprises 93 students classified as “Needs Support” and 11 classified as “Repeat.” The overall promotion rate is 89.54%, while the non-promotion rate is approximately 10.46%.

02. Assessment Performance Is the Clearest Observed Difference

Promoted students achieve an average Internal Exam score of 65.0, compared with 37.8 among non-promoted students. Their average Assignment scores are 69.9 versus 49.4, while Previous Exam scores are 66.5 versus 48.1.

The Internal Exam shows the largest average score gap, at approximately 27.2 points. These differences highlight assessment performance as an important area for further investigation and targeted academic support.

03. Attendance Alone Does Not Clearly Distinguish Outcomes

Average attendance is 75.7% among promoted students and 77.2% among non-promoted students. Across attendance bands, the non-promotion rate ranges from approximately 9.0% to 11.8%, without a consistent downward trend as attendance increases.

In this dataset, attendance alone does not clearly distinguish students by promotion outcome. Attendance should therefore be considered alongside assessment scores and other academic indicators rather than used as the sole indicator of academic risk.

04. Course-Wise Retention Risk Varies

BCA has the highest observed non-promotion rate at approximately 13.79%, followed by B.Sc Mathematics at 13.64% and BBA at 12.20%. BA Economics has the lowest observed rate at approximately 6.47%.

These differences provide a useful basis for prioritising further review. They do not, by themselves, establish that a course is weaker or that its teaching quality is lower. Course-level assessment patterns and other relevant factors should also be examined.

05. Student Distribution Across Departments and Study Years

Science is the largest department, with 367 students, representing approximately 36.92% of the dataset. Humanities accounts for 242 students (24.35%), Commerce for 146 (14.69%), Management for 123 (12.37%), and Computer Applications for 116 (11.67%).

By year of study, Year 2 has the largest student group, with 358 students (36.02%), followed by Year 1 with 322 (32.39%) and Year 3 with 314 (31.59%). These distributions provide context for interpreting the report's performance comparisons.

06. Early-Warning Analysis Identifies a Higher-Risk Subgroup

Using the rule that flags students scoring below 50 in both Internal Exams and Assignments, 68 students are identified for closer review. Of these, 43 are not promoted, meaning approximately 63.2% of the flagged group falls into the non-promoted category.

The rule identifies approximately 41.3% of all 104 non-promoted students. Among the 926 students not flagged by this rule, 61 are still not promoted, a rate of approximately 6.6%.This rule is useful for prioritising support, but it does not identify every student who may need assistance. It should be used alongside wider academic monitoring.

RECOMMENDED SOLUTIONS

RECOMMENDED SOLUTIONS — WHAT SHOULD HAPPEN NEXT?

01. Introduce an Early Assessment Checkpoint
Identify students scoring below 50 in both Internal Exams and Assignments. The current warning rule flags 68 students, of whom 43 are not promoted. Use this group to prioritise early academic review rather than waiting until the final examination period.
However, because the rule does not capture every non-promoted student, it should complement broader academic monitoring rather than determine support eligibility on its own.

02. Provide Targeted Academic Support
Prioritise students who demonstrate weaknesses in Internal Exams and Assignments. Introduce topic-based revision sessions, guided practice, faculty mentoring, and additional learning resources based on the specific areas in which each student struggles.
Students classified as “Needs Support” should receive timely assistance, while students classified as “Repeat” may require individual academic reviews and a more structured recovery plan.

03. Monitor Course-Wise Risk and Investigate Differences
Give closer attention to courses with comparatively higher non-promotion rates, particularly BCA, B.Sc Mathematics, and BBA in the current analysis.
Review course-level assessment performance, learning difficulties, and access to academic support before deciding on interventions. Use the course-wise rates as signals for investigation, not as standalone measures of course quality or teaching effectiveness.

04. Use Attendance as a Supporting Indicator
The current comparison does not show lower average attendance among non-promoted students. Therefore, attendance should not be treated as the only or primary trigger for academic intervention. Continue to monitor attendance, but interpret it alongside Internal Exam, Assignment, Previous Exam, and overall academic performance. This provides a more balanced basis for identifying students who may need support.

05. Monitor Progress Through Reassessment
After providing academic support, conduct follow-up assessments and compare students' scores with their initial results. Track changes in Internal Exam performance, Assignment scores, non-promotion rates, and the number of students requiring continued assistance.
Use these results to evaluate whether the interventions are helping and to identify where additional support is needed.

06. Establish a Regular Academic Review Process
Review student performance and course-wise outcomes at regular intervals. Use the dashboard filters to examine differences across departments, courses, genders, states, and years of study.
Document the interventions implemented, the students reached, and subsequent academic outcomes. This will help the institution make future support decisions using observed results rather than assumptions.

Expected Benefits
These recommendations provide a structured approach to identifying academic risk, prioritising support, monitoring progress, and reviewing differences across courses and student groups. Their actual effectiveness should be evaluated using subsequent assessment results and promotion outcomes rather than assumed in advance.